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Project Name: OpenCart

Project Functional Requirement Specification

(Version 1)

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1. Document Information

Documentation Information	Details
Project Name	OpenCart E-Commerce Appkication
Document Name	Functional Requirement Specification(FRS)
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Application Type	E-Commerce Web Application
Testing Scope	Customer
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Document Status	Final
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2. Purpose of Documentation

The purpose of this Functional Requirement Specification (FRS) is to document the functional requirements of the OpenCart e-commerce application. This document provides a clear reference for understanding the system functionality and serves as a basis for designing test scenarios and test cases.

3. Project Overview

OpenCart is an e-commerce web application that provides an online platform for customers to browse and purchase products. The application includes various customer-facing functionalities such as user registration, login, product search, product display, product comparison, wishlist, shopping cart, checkout, and order placement.

The application also provides a Backend/Admin Panel that allows authorized administrators to manage customers, products, categories, inventory, orders, and reports.

This project focuses on performing manual testing of both the Customer Frontend and Admin Backend to verify that the application's functionalities work according to the defined functional requirements. The testing activities include test scenario and test case preparation, functional testing, positive and negative testing, defect identification and reporting, retesting, regression testing, and end-to-end testing.

The FRS document defines the functional requirements of the OpenCart application and serves as a reference for preparing test scenarios, test cases, RTM, and other QA documentation.

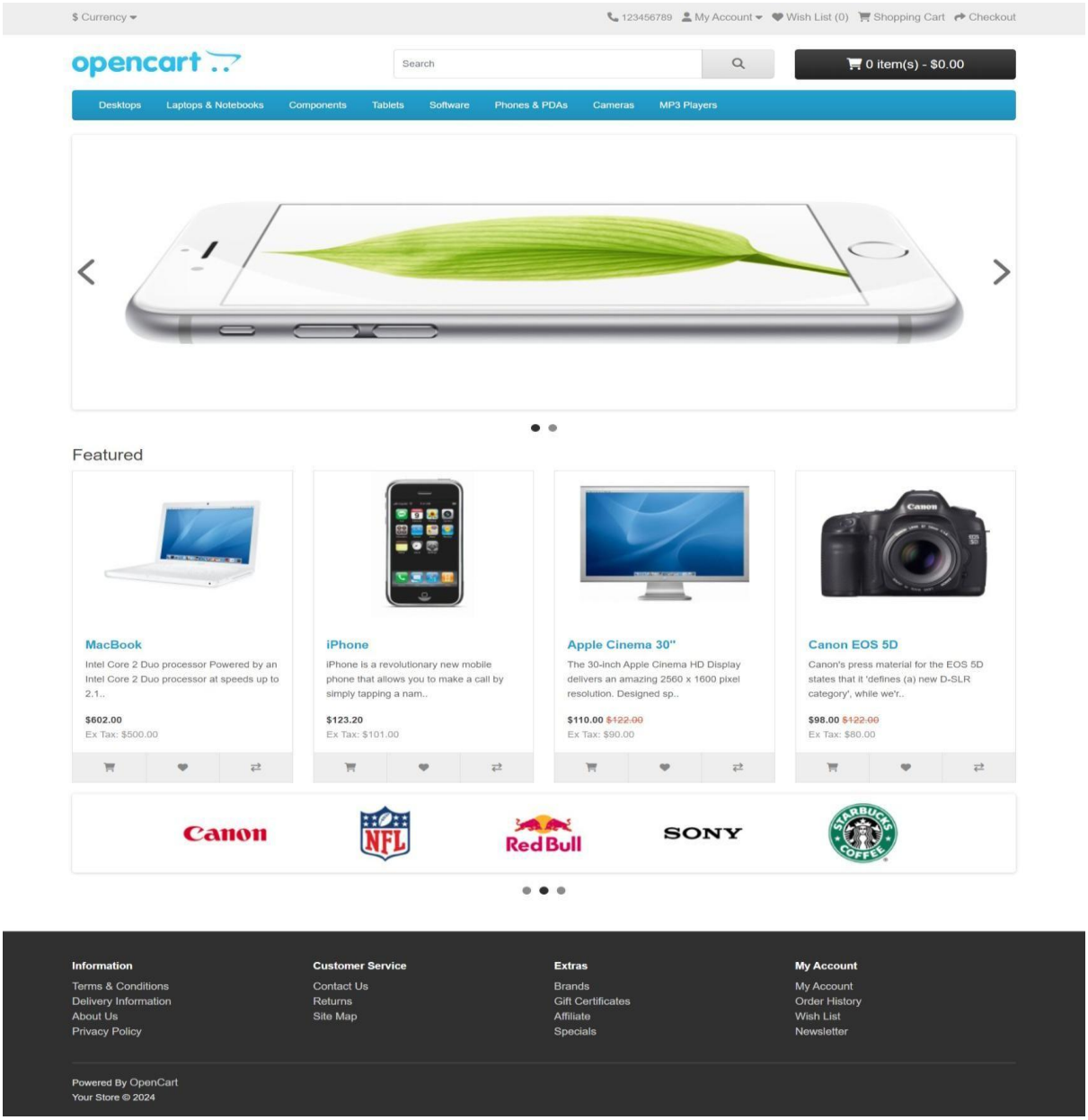


Fig.1

4. Project Scope

The scope of this project is to manually test the OpenCart e-commerce application by validating its major customer-facing and administrative functionalities. The testing covers user authentication, product-related functionality, shopping cart, checkout, order processing, customer management, product management, inventory, and reporting. The project also includes test case design, test execution, defect reporting, retesting, regression testing, and end-to-end validation to ensure that the application functions according to the defined requirements.

5. Functional Requirement

The functional requirements define the features and expected behavior of the OpenCart e-commerce application. The requirements cover both the Customer/Frontend and Admin/Backend modules.

5.1 Customer Requirements

Registration

The system shall allow new customers to create an account by providing the required registration information.

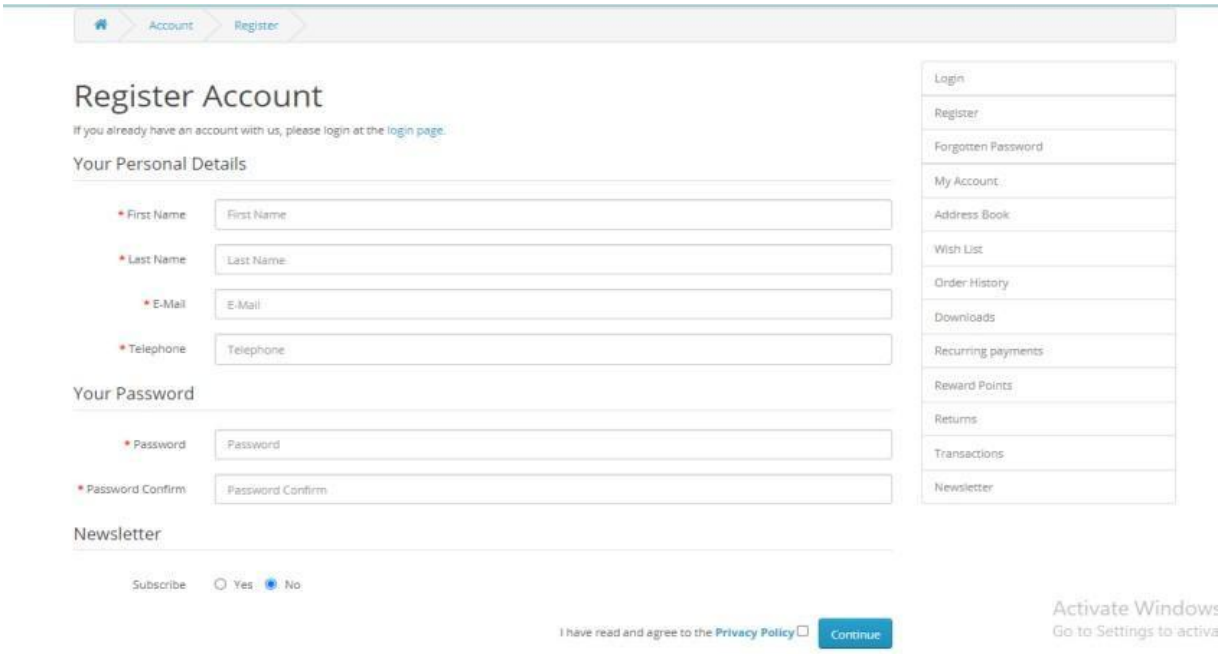


Figure 13

Login

The system shall allow registered customers to securely log in using valid credentials.

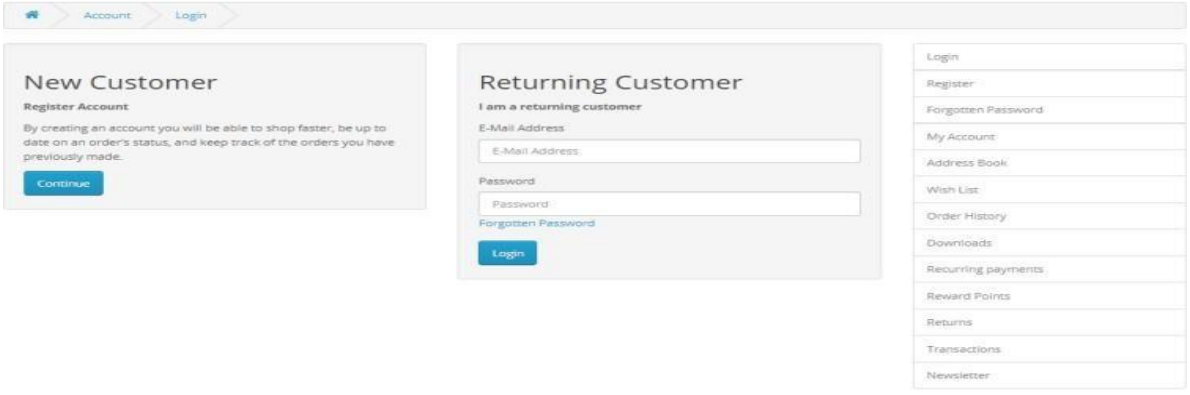


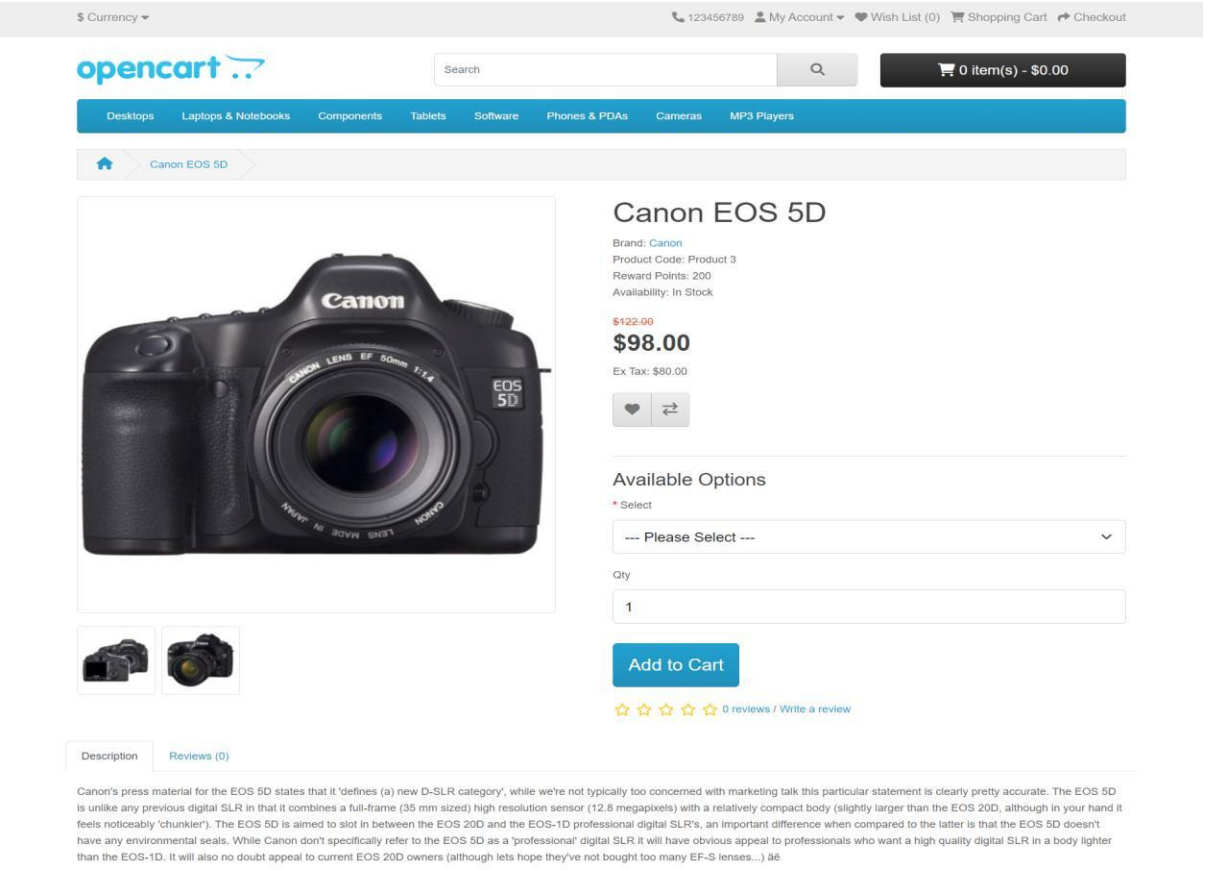
Figure 14

Logout
The system shall allow authenticated customers to log out and terminate their active session.

ForgetPassword
The system shall allow registered customers to initiate password recovery using their registered email address.

ProductSearch
The system shall allow customers to search for products using the available search functionality.

ProductDisplay
The system shall display relevant product information such as product name, image, description, price, and availability.



ProductCompare

The system shall allow customers to add products to a comparison list and compare available product information.

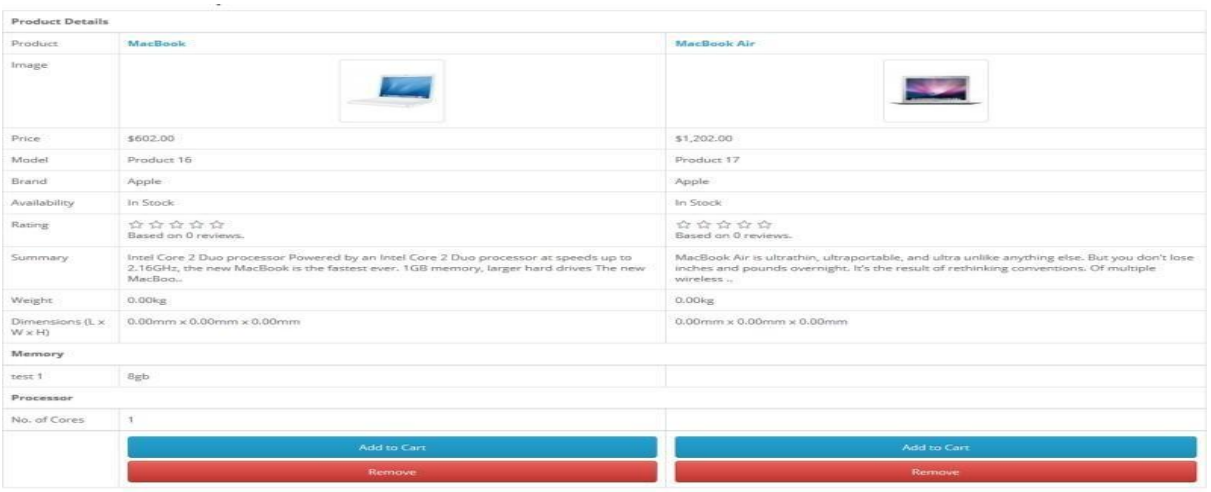


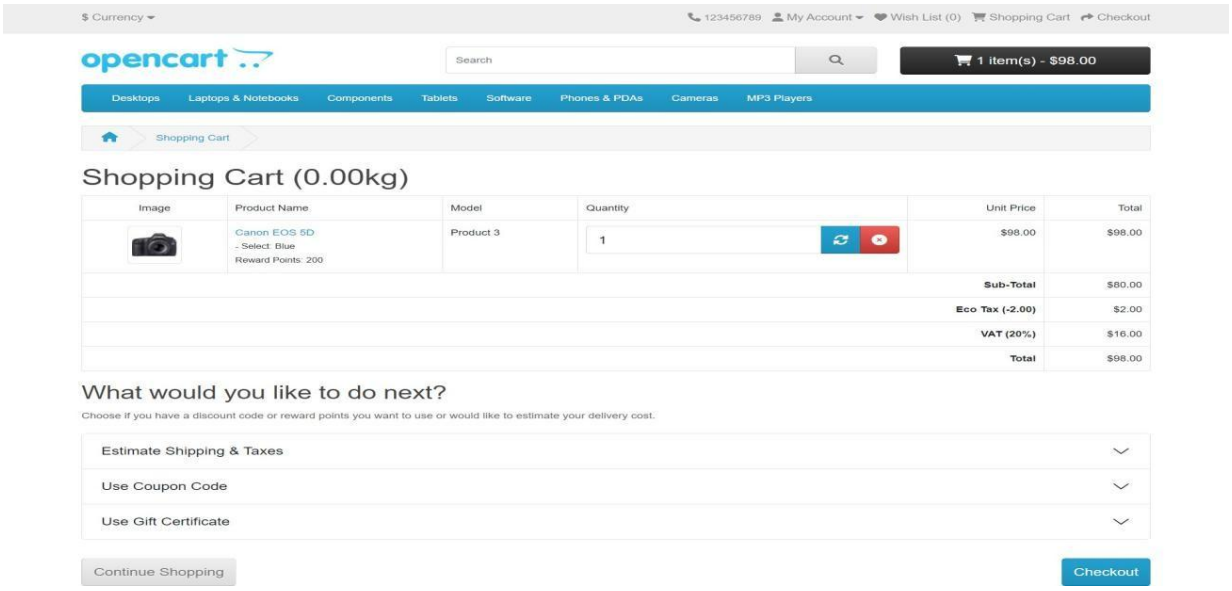
Figure 9

Wishlist

The system shall allow registered customers to add, view, and remove products from their wishlist.

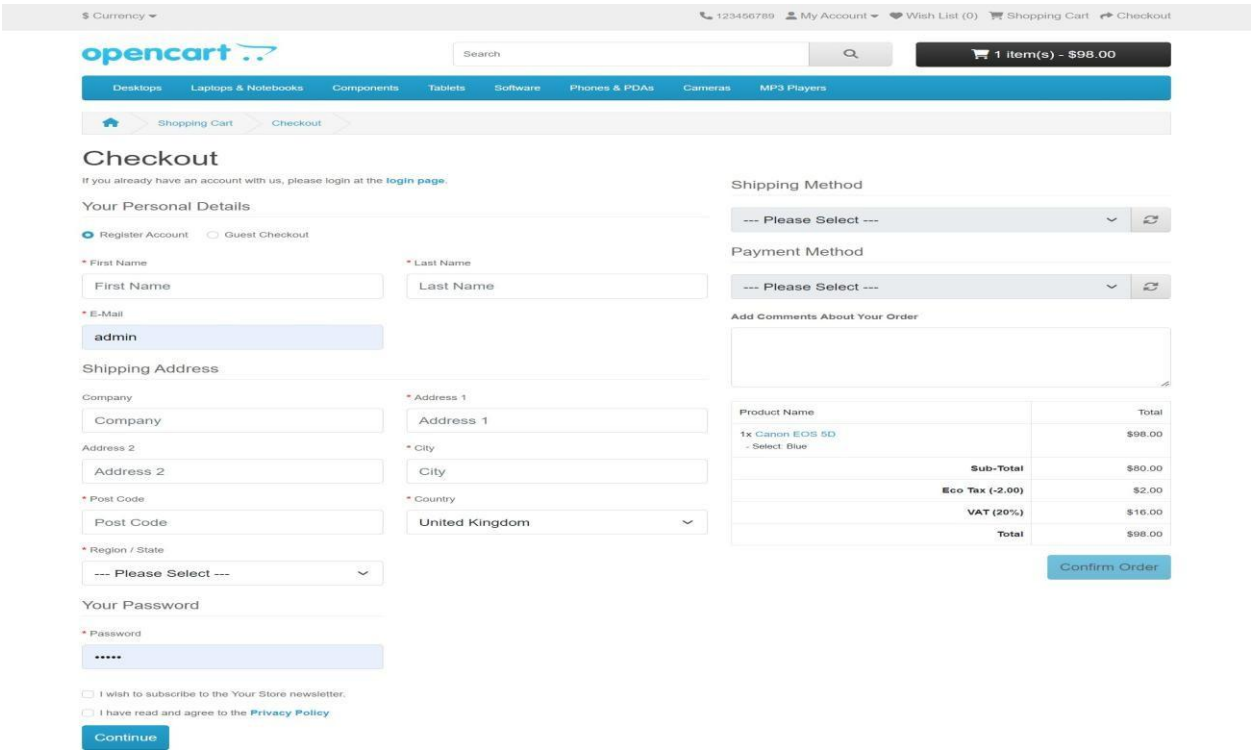
ShoppingCart

The system shall allow customers to add, update, and remove products from the shopping cart.



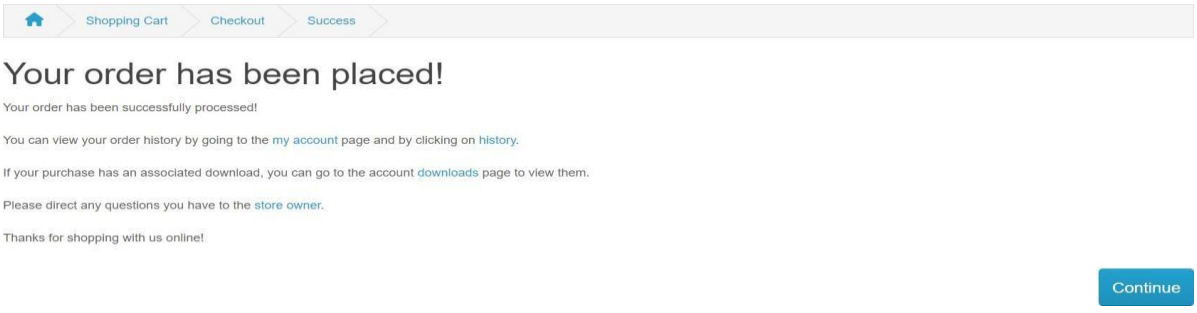
Checkout

The system shall allow customers to complete the available checkout process by providing the required billing, delivery, shipping, and payment information.



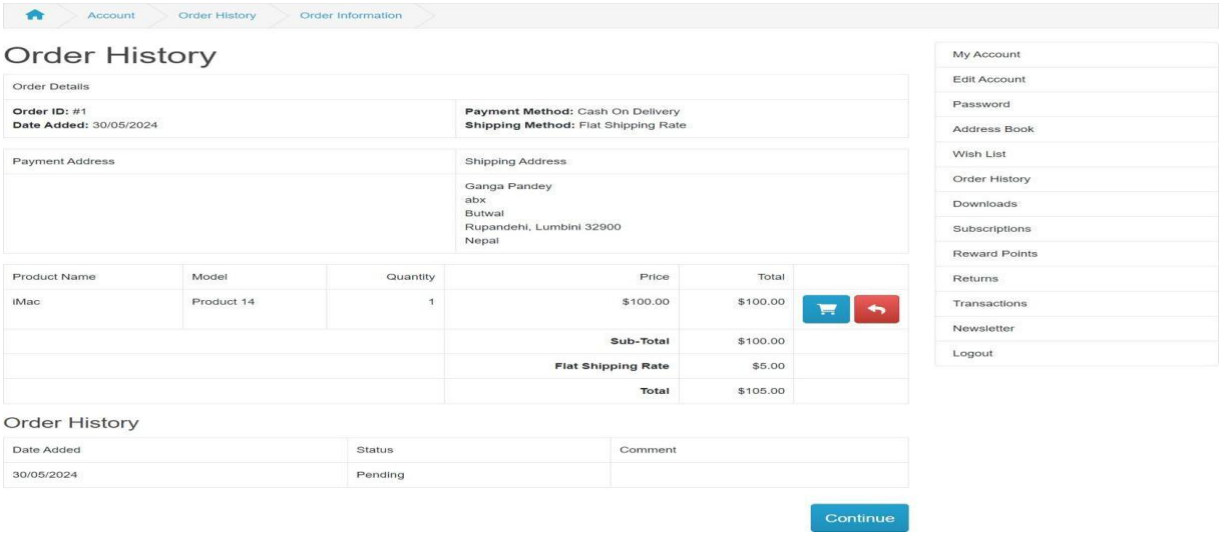
OrderPlacement

The system shall allow customers to review and place an order after completing the required checkout information.



OrderHistory

The system shall allow authenticated customers to view their available order history and order details.



Home page
The home page is arguably the most important page in the shop, in terms of presentation. In most cases, this will be the first page that a customer interacts with (especially if they are directed to the store site from a search engine). The shop's homepage needs to be user-friendly, while at the same time highlighting the shop's products.
The first step in becoming familiar with the storefront is understanding the anatomy of the Open Cart default homepage.

The header
The header will be displayed at the top of the page, on every page of the store; not just the home page.
The header has the following navigation options:

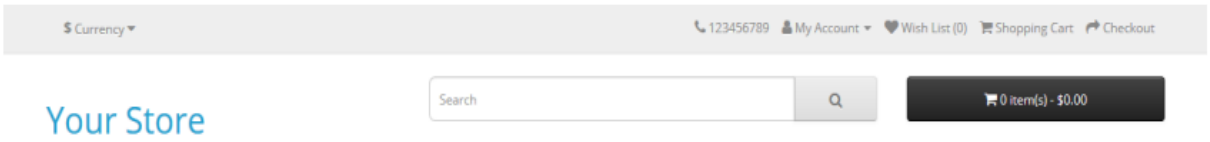


Figure 2

● **Store Logo:** Clicking on this logo will direct the customer back to the home page of the store.

● **Currency Block:** The customer can select which currency the store's products will be in by clicking on any of the currency icons.

● **Shopping Cart:** Displays the number of items purchased, and the total price of the order. Clicking on the button will contain all of the products added to the cart and an option to "View Cart" or "Checkout".

● **Search box:** The customers can type in the search box to search for a product within the store's product categories.

● **Links:** Links the customer to the Home page, Wish List, My Account, Shopping Cart, and Checkout.

● **Telephone**: Company telephone number.

● **My Account:** Customers can register or log in from here.

The top menu
The top menu category only displays the top parent categories of products. See Categories for more information on how to create and assign product categories. Like the header, the top menu will be displayed on every page. When the customer's mouse is dragged over a category, a drop-down menu will display the sub-categories for that parent category. When a parent category is clicked, the customer will be directed to the category page, which displays all the products within that category.



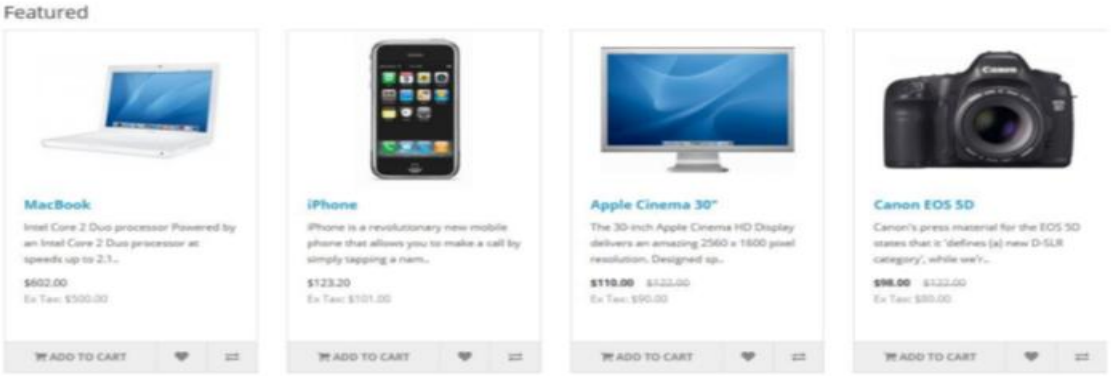
Figure 3

Slideshow
The slideshow displays several product banners of your choice by alternating the images in a slideshow. After a certain amount of time, one banner will shift to the next banner. Banners in this slideshow are useful for highlighting certain products to be easily accessible by the customer. When the banner is clicked on, the customer will be directed to the product on the banner's page.



Unlike the top menu and header, the slideshow in the OpenCart default can only be viewed on the home page in this position.

Featured products
OpenCart gives you the option of featuring specific products of their choosing on the home page.

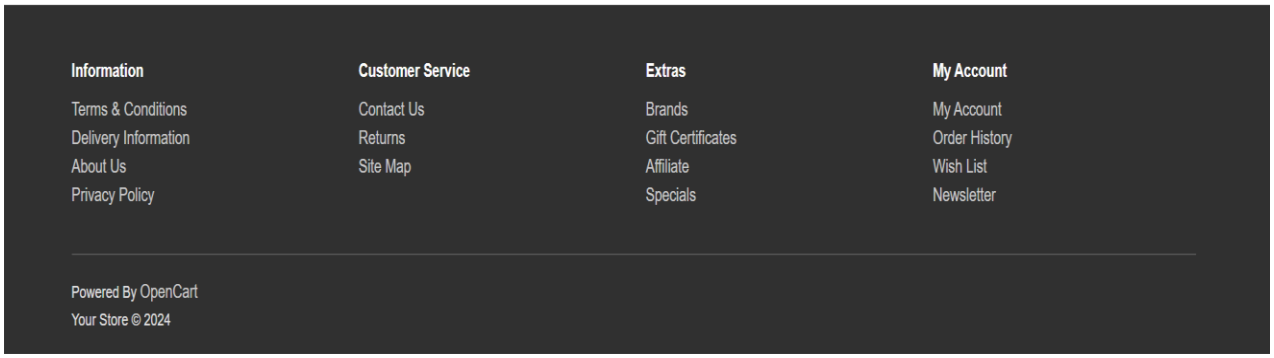


Footer

The footer is located at the bottom of every page, not just the Home Page. This block of miscellaneous links is useful in sorting relevant pages for the customer that may not be logically sorted anywhere else.

The organizational scheme of the footer can be divided into the following sections:

- **Information:** “About Us”, “Delivery Information”, “Privacy Policy”, “Terms & Conditions”
- **Customer Service:** "Contact Us", "Returns", "Site Map"
- **Extras:** "Brands", "Gift Vouchers", "Affiliates", "Specials"
- **My Account:** "My Account", "Order History", "Wish List", "Newsletter”



6. Information Architecture

The OpenCart application is divided into two primary areas: the Customer/Frontend and Admin/Backend. The Customer/Frontend provides product browsing, account management, shopping cart, checkout, and order-related functionality. The Admin/Backend provides authorized administrators with access to dashboard, customer, product, category, inventory, order, and reporting functionalities. This structure defines the major navigation and functional areas covered by the manual testing project.

7. Business Rules / Validation Requirements

The OpenCart application shall validate mandatory fields, customer credentials, email formats, passwords, product information, quantities, checkout details, and administrative operations. Only authorized users shall be permitted to access protected customer or admin functionality. The system shall maintain accurate product, inventory, cart, order, and report information and shall provide appropriate validation or error messages for invalid input.

8. Assumptions

The OpenCart application, test environment, required customer and administrator credentials, test data, supported browsers, and required application dependencies are assumed to be available and accessible during testing. It is assumed that the documented functional requirements represent the expected system behavior and that required application fixes will be provided for retesting. Testing will be performed within the defined project scope and according to the agreed testing process.

9. Constraints

The OpenCart e-commerce system is subject to certain technical, operational, security, and environmental constraints. The application depends on a compatible web server, database, supported browser, and stable internet connection for proper operation. System performance may vary depending on server resources, database size, network speed, and the number of concurrent users. Payment, shipping, email, SMS, and other third-party services depend on the availability and compatibility of external APIs or service providers. Access to administrative and restricted functionalities must be controlled through proper authentication and authorization. The system must also comply with configured business rules related to product availability, customer information, order processing, inventory, payment, and delivery. Compatibility may be affected by the versions of OpenCart, operating system, browser, plugins, and extensions used. Regular maintenance, security updates, database backups, and system upgrades may require temporary service interruption. Testing and system behavior may also vary depending on the configured test environment, browser, network conditions, database configuration, and availability of external services

10. Risks

The OpenCart e-commerce system may be affected by various risks during development, testing, deployment, and maintenance. These risks may include security vulnerabilities, unauthorized access, data loss, payment gateway failures, third-party service unavailability, server or database failures, network issues, incorrect inventory information, and unexpected application errors. Changes or updates to OpenCart, extensions, plugins, or third-party services may also cause compatibility issues or affect existing functionality. High traffic or a large number of concurrent users may impact system performance. To minimize these risks, appropriate security controls, regular backups, data validation, error handling, monitoring, compatibility testing, and regression testing should be performed throughout the project lifecycle.

11. Acceptance Criteria

The OpenCart e-commerce system shall be considered acceptable when all specified functional and non-functional requirements defined in this FRS have been implemented and successfully verified. Users should be able to register, log in, search and view products, manage the shopping cart and wishlist, place orders, make payments through the configured payment method, and view relevant order information. Administrators should be able to manage products, categories, customers, orders, inventory, and reports according to their assigned permissions. All critical and high-severity defects should be resolved or formally accepted before release. The system should also meet the defined security, usability, compatibility, performance, and reliability requirements and should successfully pass the planned testing activities.

12. Sign-Off

The following parties have read and agree with this Requirements Definition document for the OpenCart application account module functionality.

After approval of this Requirements Definition phase, any significant changes in the scope of this project will require validation of existing project costs and schedules.

Name:

Business Lead

Date:

Name:

Project Manager

Date:
